BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 298 · Backtracking in computational solutions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When does a recursive call stop?

Q2. What is a search tree node?

Q3. Why may pruning reduce runtime?

Q4. Insert 12 into BST root10 with right child15 and left child5. Show the comparisons and position.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

backtracking — Explore choices and undo a failed path.

pruning — Discard an impossible branch.

search tree — Branches of possible decisions.

base case — Condition ending a recursive branch.

New learning

Make a decision — At each position choose the current item or skip it. Track index and running total.

Reject a dead end — When the sum exceeds the target (non-negative inputs) or the list ends, undo the choice.

Backtrack — Try the other branch; success means a valid selection has been found. Exponential worst-case growth remains possible.

Worked example

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

1. Predict first — The first path over-shoots: 2+3+4=9, so reject that branch.

2. Explain the method — Undo 4, skip 3, then include 4 → 6.

We Do · Practise with guidance

Find a subset totalling 5 from [1,3,4] using inclusion/exclusion.

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Include 1 and 3 →4
Include 4 →8 reject
Undo 4; exclude 4 → dead end at 4
Backtrack: skip 3; include 4 →5 success
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Predict first

Including 1 and 3 produces 4; adding 4 overshoots 5.

Explain the method

Undo 4, skip 3, then include 4 → 5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain choice, dead end and backtrack using [2,3,4] target6. [6]

Question 2

Write a recursive include/exclude Boolean procedure for subset sum with non-negative integers, including base cases. [8]

Question 3

Explain why backtracking is useful for 4 queens but not guaranteed polynomial time. Give one safe-placement rule. [6]

Exit Ticket · Answer before checking

Backtracking can require trying an earlier alternative after a dead end.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. When a base case is met.

Q2. A partial choice/state considered in the search.

Q3. It avoids exploring states proven unable to succeed.

Q4. 12>10 ⇒ go right to15; 12<15 ⇒ insert as left child of15.

You Do pages · Independent questions

Q1. Try subset [2,3,4] [1], sum9 [1], prune as >6 for non-negative inputs [1], undo last choice [1], exclude 3 [1], choose 4 with 2 for sum6 [1].

Q2. Check total==target return true [2]; end-of-input false [1]; prune total>target [1]; include branch [2]; exclude branch [1]; combine with OR [1].

Q3. Rows/columns/diagonals constrained [2]; choose next row/column then recurse [1]; undo when no legal square [1]; worst-case combinatorial branching [1]; safe placement checks no shared column or diagonal [1].

Exit Ticket · Model answer

True; reversing a choice lets the search explore another branch.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding